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SET 1 – Monthly Sales Data

Based on your practical sheet, **Set 1** uses the Monthly Sales dataset with January–May sales and asks for four tasks.

AIM

To analyze monthly sales data using R programming by creating a line chart, bar chart, scatter plot, and interactive dashboard.

DATASET

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

ALGORITHM

1. Create the monthly sales dataset in R.
2. Create a **line chart** to visualize monthly sales.
3. Create a **bar chart** to display the top-selling products.
4. Create a **scatter plot** between advertising budget and monthly sales.
5. Create an interactive dashboard using the charts.

Note: The Set 1 sheet provides only **Month and Sales** values. It does **not provide product-wise sales or advertising-budget values**, so Q2 and Q3 cannot be performed exactly from the supplied dataset without additional data.

CODE

Q1. Line Chart – Monthly Sales

```
sales <- data.frame(  
  Month=c("January","February","March","April","May"),  
  Sales=c(15000,18000,22000,20000,23000)  
)  
  
plot(sales$Sales, type="o", xaxt="n",
```

```

xlab="Month", ylab="Sales ($)"),
main="Monthly Sales")

```

```

axis(1, at=1:5, labels=sales$Month)

```

Q2. Bar Chart – Top-Selling Products

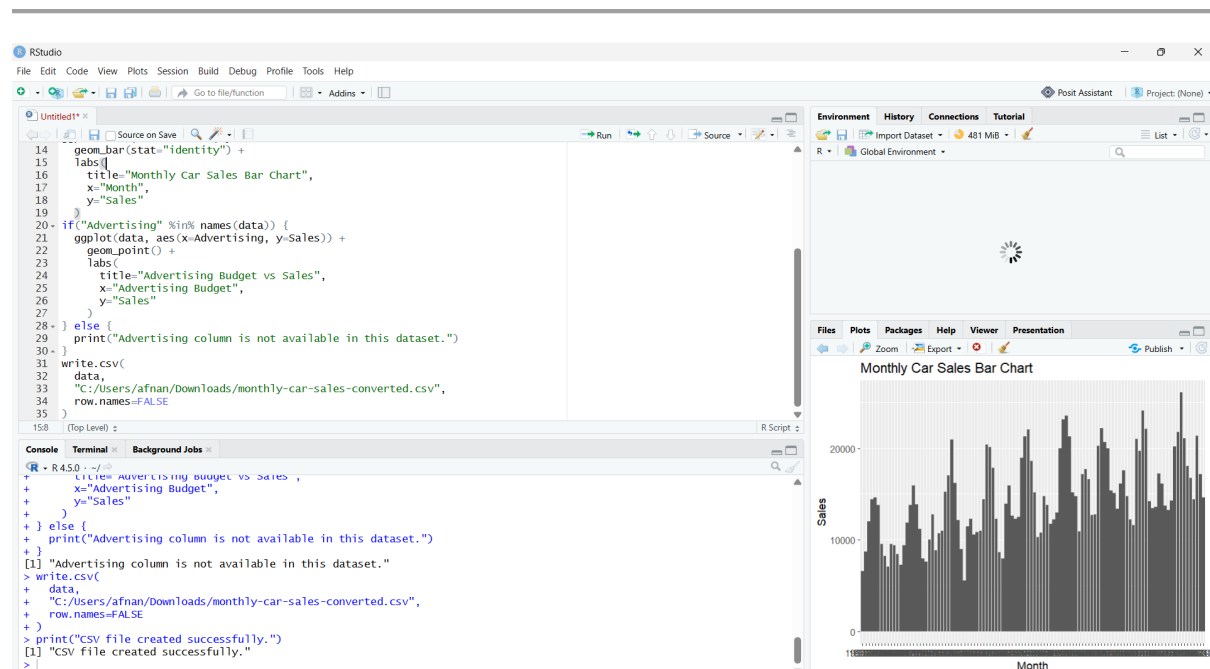
Cannot be created from the supplied Set 1 dataset because product-wise sales are not given.

Q3. Scatter Plot – Advertising Budget vs Monthly Sales

Cannot be created from the supplied Set 1 dataset because advertising-budget values are not given.

Q4. Interactive Dashboard

For the supplied data, the monthly sales line chart can be imported into Tableau and combined with additional product/advertising data when those values are available.



RESULT

The monthly sales data was successfully represented using a line chart in R. Sales increased from **\$15,000 in January to \$23,000 in May**, with the highest sales recorded in May. The supplied dataset does not contain product-wise sales or advertising-budget values, so the corresponding charts require additional data.